

# The Effect of School Choice on Participants: Evidence from Randomized Lotteries

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# Background

Does winning a randomized lottery for admission to schools that are better in a number of dimensions improve student outcomes?

## Why does this topic matter?

- Increasing the scope of schooling alternatives → global
- To avoid the issue of nonrandom attendance

## Treatment:

- Winning admission to a preferred Chicago Public Schools (CPS)

## Key outcomes:

- Traditional
- Nontraditional

# Motivation: Evaluation Problem

**The rise of choice schools in Chicago**

→ seeking alternatives that align with students' needs

**The selection bias trap: Students who choose better schools may also have:**

- More motivation
- More involved parents
- More achievements beforehand, etc

→ Cannot identify the true causal effect

Observed Difference  
= School Effect + Student Differences

**A natural experiment  
randomized lotteries,  
eliminating selection bias  
and producing a reliable  
estimate of the true cause  
and effect**

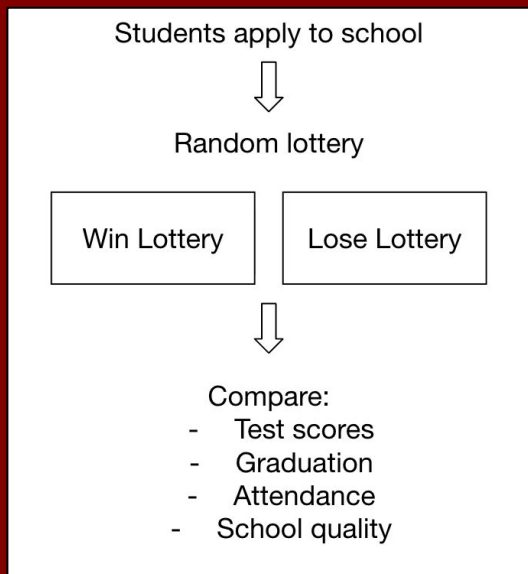
# Empirical design

## Problem:

School choice suffers from selection bias

## Solution:

Randomized admission lotteries



## Identifying Assumptions:

-Winners and losers are identical **on average**

-Only systematic difference is lottery outcome

-Therefore differences in outcomes can be interpreted causally

# Empirics: Difference in Mean Estimator

$$\delta_j = E[Y_i | \text{Win}_{ij} = 1; \text{Apply}_{ij} = 1] - E[Y_i | \text{Win}_{ij} = 0; \text{Apply}_{ij} = 1],$$

where

- $Y_i$  = outcome for student  $i$
- $\text{Win}_{ij} = 1$  if student  $i$  won lottery  $j$ , 0 otherwise
- $\text{Apply}_{ij} = 1$  if student applied to lottery  $j$ , 0 otherwise
- $\delta_j$  = average difference between winners and losers within the same lottery

Lottery assignment is random

→ No selection bias

→ Difference in means of observed student outcomes between students who win and lose the lottery gives a consistent estimate of the **causal effect of winning the lottery**

$$\hat{\tau}_{RCT} = \bar{Y}_{\text{Winner}} - \bar{Y}_{\text{Loser}}$$

# Empirics: Regression Estimator

$$Y_i = \delta(\text{Win\_Lottery}_{ia}) + \Gamma(\text{Lottery}_a) + e_{ia},$$

$$Y_i = \beta_0 + \beta_1 \text{Win}_i + \text{Lottery Fixed Effects} + X_i + \varepsilon_i$$

where

- $Y_i$ : student outcome
- $\text{Win}_i$ : won lottery
- Lottery fixed effects compare students in the same lottery
- $X_i$  = demographics and prior achievement

Regression improves precision and preserves the randomized comparison at the same time.

# Data

## Data Sources & Features

- Administrative Records
- Lottery Details
- Survey Data (subset)

Applications:  
19,520

Students: 14,434

Lotteries: 194

High Schools: 19

# ITT

Lottery assignment is random

→ Winning the lottery is **independent** of students' observed and unobserved characteristics

Therefore, differences in outcomes between lottery winners and losers can be interpreted as **causal effects**.

## Intent-to-Treat (ITT)

Students are randomly assigned to win the lottery, not necessarily to attend the school

$$\text{ITT} = E[Y|\text{Win}] - E[Y|\text{Lose}]$$

# LATE

Lottery assignment can also be used as an **instrument** to estimate the **Local Average Treatment Effect (LATE)**

$$LATE = \frac{E[Y|Win=1] - E[Y|Win=0]}{E[Attend|Win=1] - E[Attend|Win=0]}$$

**where:**

Y: student outcome

Win: 1 for winning the lottery, 0 for losing

Attend: 1 for actually attending the lottery school, 0 otherwise

→ Only applies to **compliers**: students who enroll if they win the lottery and do not enroll if they lose

$$\tau_{LATE} = \frac{E(Y|Z=1) - E(Y|Z=0)}{E(D|Z=1) - E(D|Z=0)}$$

# Table IV

TABLE IV  
THE IMPACT OF WINNING A LOTTERY ON THE CHARACTERISTICS OF SCHOOL  
ATTENDED—BY LOTTERY SCHOOL TYPE<sup>a</sup>

| Dependent Variable                                  | Any School<br>(1)              | High-<br>Achieving<br>School<br>(2) | High Value<br>Added<br>School<br>(3) | High<br>Popularity<br>School<br>(4) |
|-----------------------------------------------------|--------------------------------|-------------------------------------|--------------------------------------|-------------------------------------|
| <i>Type of high school attended</i>                 |                                |                                     |                                      |                                     |
| School for which lottery applies                    | 0.280**<br>(0.012)<br>[0.069]  | 0.338**<br>(0.022)<br>[0.082]       | 0.242**<br>(0.023)<br>[0.072]        | 0.381**<br>(0.024)<br>[0.082]       |
| School in top quartile in terms of peer achievement | 0.063**<br>(0.011)<br>[0.350]  | 0.283**<br>(0.019)<br>[0.431]       | 0.174**<br>(0.020)<br>[0.422]        | 0.297**<br>(0.022)<br>[0.407]       |
| Fraction of students receiving free lunch           | -0.021**<br>(0.003)<br>[0.786] | -0.044**<br>(0.006)<br>[0.747]      | -0.027**<br>(0.007)<br>[0.736]       | -0.056**<br>(0.007)<br>[0.744]      |
| Graduation rate                                     | 0.018**<br>(0.003)<br>[0.689]  | 0.033**<br>(0.004)<br>[0.714]       | 0.023**<br>(0.004)<br>[0.715]        | 0.043**<br>(0.005)<br>[0.709]       |

All coefficients are statistically significant at 5% level

Winning the lottery actually changes where students go and treatment with school environments.

# Results

Students attended objectively better schools,

**but their academic outcomes changed very little**

| Academic Outcomes | Behavioral Outcomes            |
|-------------------|--------------------------------|
| ✗ Test scores     | ✓ Fewer disciplinary incidents |
| ✗ Graduation      | ✓ Lower arrest rates           |
| ✗ Attendance      | ✓ Lower incarceration rates    |
| ✗ Credits earned  |                                |

# Table V

TABLE V  
THE IMPACT OF WINNING A LOTTERY ON TRADITIONAL MEASURES OF STUDENT  
OUTCOMES—BY LOTTERY SCHOOL TYPE<sup>a</sup>

| Dependent Variable                    | The Effect of Winning a Lottery to |                                     |                                      |                                     |
|---------------------------------------|------------------------------------|-------------------------------------|--------------------------------------|-------------------------------------|
|                                       | Any School<br>(1)                  | High-<br>Achieving<br>School<br>(2) | High Value<br>Added<br>School<br>(3) | High<br>Popularity<br>School<br>(4) |
| <i>Outcomes at the end of 4 years</i> |                                    |                                     |                                      |                                     |
| Graduated <sup>b</sup>                | −0.044**<br>(0.018)<br>[0.583]     | −0.017<br>(0.034)<br>[0.636]        | 0.011<br>(0.051)<br>[0.635]          | −0.046<br>(0.044)<br>[0.621]        |
| <i>9th grade outcomes</i>             |                                    |                                     |                                      |                                     |
| Reading percentile score              | −0.013**<br>(0.005)<br>[0.415]     | −0.009<br>(0.007)<br>[0.469]        | −0.010<br>(0.008)<br>[0.487]         | −0.010<br>(0.009)<br>[0.456]        |
| Algebra end-of-course exam score      | −0.002<br>(0.007)<br>[0.446]       | −0.009<br>(0.010)<br>[0.492]        | −0.005<br>(0.012)<br>[0.518]         | −0.008<br>(0.012)<br>[0.488]        |
| English I end-of-course exam score    | −0.001<br>(0.004)<br>[0.596]       | 0.000<br>(0.005)<br>[0.620]         | 0.013*<br>(0.006)<br>[0.628]         | −0.002<br>(0.005)<br>[0.619]        |

no statistically significant improvements in most academic outcomes.

Nearly all coefficients are statistically insignificant at 5% level.

# Validity

## Internal Validity

- Random assignment
- Comparable treatment & control groups
- Low attrition
- Robust results

## External Validity

Generalizes well to:

Urban public school systems  
Lottery-based admissions

Less generalizable to:

Rural districts  
Voucher programs  
Students who never applied

# Critique: Limitations

The study estimates the effect of winning a Chicago school choice lottery, so its findings may not fully generalize to other school choice programs or different educational systems.

→ Strong **internal validity**, but limited **external validity**

## Selective attrition

- Only students who are still enrolled in CPS are observed
- Students who leave CPS cannot be followed

→ The estimated effects could be biased if lottery winners and losers leave at different frequencies, and the students who leave are systematically different

# Overall Conclusion

## Results:

- Better schools  $\neq$  better test scores
- Some behavioral improvements

## In design:

- Strong causal design
- Important implications for making policies

# Citation

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Cullen, Julie Berry, et al. “The Effect of School Choice on Participants: Evidence from Randomized Lotteries.”

*Econometrica*, vol. 74, no. 5, 2006, pp. 1191–1230. *JSTOR*, [www.jstor.org/stable/3805923](http://www.jstor.org/stable/3805923),

<https://doi.org/10.2307/3805923>.